The empirical recurrence for OEIS A283787: recovering a conjecture that is not written down

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Abstract

OEIS A283787 records "Empirical recurrence of order 69", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 153$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 69, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 69 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283787 is "Number of 4Xn 0..1 arrays with no 1 equal to more than two of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly two elements.". It has offset 1 and begins

$$0, 6, 745, 10910, 169426, 2343698, 30330979, 378503754, \ldots$$

The entry states:

> Empirical recurrence of order 69 (see link above)

As of the "Last modified" line on the live entry (Mar 16 2017 12:25:03, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 721$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n\tau$, so they may be merged without changing any count; merging leaves $S' = 153$ of the 721, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 153.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 306$ exact terms of the model returns order 69 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{69} c_i\, a(n-i),$$

c_1	24	c_{19}	−196581080004	c_{37}	−2864110979625	c_{55}	2689437849
c_2	−150	c_{20}	242058678432	c_{38}	−1563940124970	c_{56}	−1131684261
c_3	86	c_{21}	978704074818	c_{39}	4202391170205	c_{57}	−2288635944
c_4	−3810	c_{22}	541753301790	c_{40}	2550678582093	c_{58}	221744691
c_5	42813	c_{23}	−1669951457376	c_{41}	212836626411	c_{59}	130560813
c_6	7973	c_{24}	−3165702622472	c_{42}	328474185739	c_{60}	87933060
c_7	−41748	c_{25}	−532972353252	c_{43}	−2037496157763	c_{61}	18632679
c_8	−4229442	c_{26}	4961393802789	c_{44}	−549968671773	c_{62}	−21053706
c_9	390265	c_{27}	5837168578130	c_{45}	355497500027	c_{63}	234999
c_{10}	35716965	c_{28}	88740196668	c_{46}	−282830782146	c_{64}	−744072
c_{11}	166605900	c_{29}	−7260650494209	c_{47}	606881815443	c_{65}	479268
c_{12}	−181261081	c_{30}	−8618954680715	c_{48}	195936561784	c_{66}	45728
c_{13}	−1959547515	c_{31}	296648475738	c_{49}	−240357829392	c_{67}	−13872
c_{14}	−1940171169	c_{32}	6696021891351	c_{50}	35379326646	c_{68}	−1920
c_{15}	9640611258	c_{33}	6880493774783	c_{51}	−44498835134	c_{69}	−64
c_{16}	30502941996	c_{34}	3721330376784	c_{52}	−34939348899		
c_{17}	−5548966947	c_{35}	−4554775283916	c_{53}	28518562596		
c_{18}	−147702028552	c_{36}	−5031773149333	c_{54}	5880145327		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 69, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $69 + 4$ terms removed returns order 69 as well, so $\deg q_{\min} = 69 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $69+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 69 that this sequence satisfies — and that family turns out to have exactly one member.

References

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